On my honor I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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1) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -3\gamma & 1 \\ 1 & \beta & 3 & \alpha^2 \\ 1 & 4\alpha & -4 & 5 \end{bmatrix}$ $\xrightarrow{-3\alpha_1}$

3) $u = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$ $|u| = 1$

$H = I - 2uu^T$

a) $H^2 = H \cdot H = (I - 2uu^T)(I - 2uu^T)$

$u^T u = [a_1\ a_2 \dots a_n] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = 1 \qquad \sum_{i=1}^{n} a_i^2 = 1$

$H^2 = I^2 - 2I(uu^T) - 2(uu^T)I + 4(u^T u)^2$ since $uu^T = 1$

$= I^2 - 2uu^T - 2uu^T + 4uu^T$

$= I - 4uu^T + 4uu^T = I$ $\boxed{H^2 = I}$

b) H is a symmetric matrix so $H = H^T$

$H \cdot H = H \cdot H^T$

$H^2 = H \cdot H^T$ since $H^2 = I$

$I = H \cdot H^T$

$H H^T = I$ it proves that H is an orthonormal matrix

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_{k+1} \end{bmatrix}$$ because is proportional to the difference

$$\begin{vmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{vmatrix} \quad (-2-\lambda)^2 - 1 = 0$$

$\lambda^2 + 4\lambda + 3 = 0$ $\quad \lambda = -1$ $\quad \lambda = -3$

$\lambda \quad -1$

$\lambda \quad -3$

$\lambda_1 = -1$ $\quad \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-x_1 + x_2 = 0$

$x_1 = t$

$x_2 = t$ $\quad + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = V_1$

$\lambda_2 = -3$ $\quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$x_1 + x_2 = 0$

$x_1 = -t$

$x_2 = t$ $\quad + \begin{bmatrix} -1 \\ 1 \end{bmatrix} = V_2$

$C_1 \lambda_1 V_1 + C_2 \lambda_1 V_2$

$C_1 (-1)^n \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 (-3)^n \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$n = 0$ $\quad \begin{bmatrix} V_0 \\ W_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} = C_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$C_1 - C_2 = 1$ $\quad C_1 = 0{,}5$

$C_1 + C_2 = 0$ $\quad C_2 = -0{,}5$

$0{,}5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0{,}5 \begin{bmatrix} -1 \\ 1 \end{bmatrix} \cdot (-3)^n$

$n = 1$

$(0{,}5) \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0{,}5 \begin{bmatrix} -1 \\ 1 \end{bmatrix} \cdot (-3)$

$\begin{bmatrix} -0{,}5 \\ -0{,}5 \end{bmatrix} - \begin{bmatrix} -1{,}5 \\ 1{,}5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$